MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RELAY, 4PDT, ANALOG, KEY CODE NRN

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for procurement of an electronic 4PDT relay module.

1.2 Module description. The following electrical parameters reflect the 0° to end-of-life limits from table I.

Name	Relay, 4PDT
Family	Analog
Key code	NRN
Size	1F
Weight	283.5 grams maximum
Failure rate	0.301 failures/10^6 hours maximum plus 0.100 failures/10^6 contact operations
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	11.5 W
Operate time (including contact bounce)	10 ms maximum
Release time (including contact bounce)	15 ms maximum
Pickup voltage	8 Vdc maximum
Dropout voltage	0.5 Vdc minimum
Coil resistance	64.8Ω maximum
Contact current	10 A maximum (resistive load)
Input coil power	6.65 W maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Module function. This module contains a four-pole, double-throw, break-before-ake relay with high current and voltage switching capability. Relay pull-in time is educed to less than 10 ms by increasing the energizing voltage. Programmable resistors ermit relay power dissipation reduction at increased voltages. Inductive transients re suppressed by programming a diode across the relay coil together with external esistors.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of nvitation for bids or request for proposal, form a part of this specification to the xtent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-26 - Resistor, Fixed, Wirewound (Power Type), General Specification for.

MIL-R-6106 - Relay, Electromagnetic, General Specification for.

MIL-R-10509 - Resistor, Fixed Film (High Stability), General Specification for.

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval ublications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

PUBLICATIONS

MILITARY

NAVORD Drawing 2465232 - Relay, Four Pole Double Throw.

(All Ordnance Data publications shall be ordered from the Commanding Officer, echnical Documents Department, Naval Ordnance Station, Louisville, KY 40214.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of IL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers nd test equipment shall be correlated in accordance with MIL-M-28787.

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* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.301 failures per million operating hours plus 0.100 failures per million contact operations when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class 1 specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be NRN.

3.5.2 Size. The module size shall be 1F.

3.5.3 Weight. The module shall have a maximum weight of 283.5 grams.

3.5.4 Contact pins. The module shall have 80 contact pins positioned A1 through A40 and B1 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A and heat dissipation fins on increments B through F.

3.5.6 Depth. The module shall have a depth of 0.210 $\pm$0.005 for increments C through F per figure 4 of MIL-STD-1389.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/133, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 Thermal design. The thermal design of the module shall be such that over the class I operating fin temperature range, as specified in MIL-M-28787, the module shall conform to the following:

a. Maximum relay coil temperature shall not exceed 150°C.
b. The diode and resistor component critical temperature (CCT) shall not be exceeded.

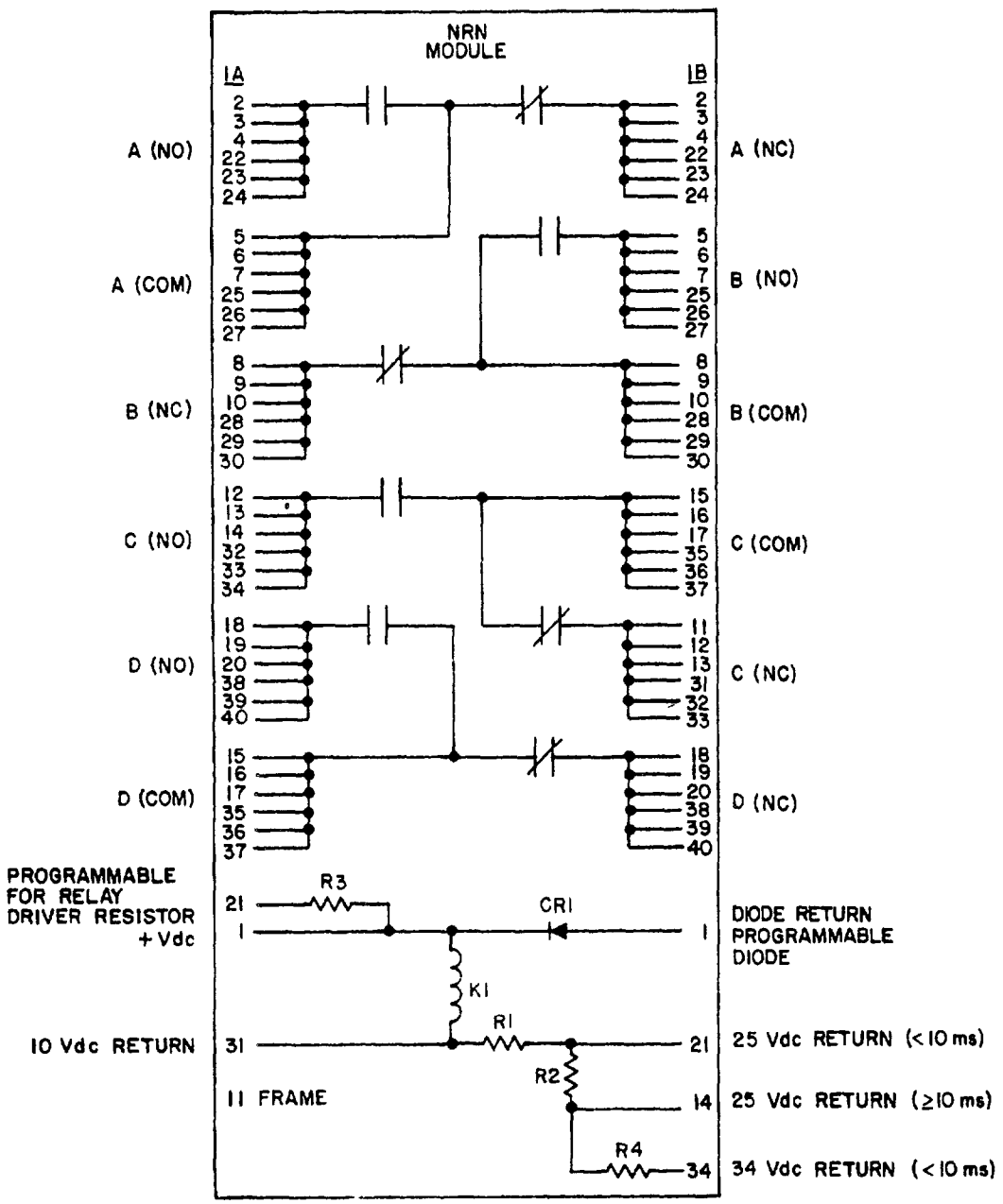
3.9 Components. The following components or their proven equivalents shall be used in the relay module.

3.9.1 Relay mechanisms. Relay mechanisms used in the module shall conform to the requirements of the relay NAVORD Drawing 2465232-2, except for dimensional and terminal requirements.

3.9.2 Resistors and diode. Resistors and a diode shall be provided as shown on figure 1. Resistance values shall be as specified on figure 1.

3.9.2.1 Wirewound resistors. The power wirewound resistors selected for use in the module shall conform to the requirements of this document and shall be in accordance with MIL-R-26, except as modified herein.

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NOTES:
1. R3 = 825Ω ±1%, 1/2 W.
2. R1 = 71.5Ω ±1%, 6.5 W.
3. R2 = R4 = 47.0Ω ±5%, 3 W.
4. CR1 = 1N647, or equivalent.

FIGURE 1. NRN module function diagram, 4 PDT relay.

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3.9.2.2 Film resistors. The film resistors selected for use in the module shall conform to the requirements of this document and shall be in accordance with MIL-R-10509, except as modified herein.

3.9.2.3 Alternate components. Alternate components may be selected provided there is documented evidence that they are proven equivalent in all functional respects and that they do not degrade the reliability requirements of this specification.

3.10 Life and contact rated load requirements. The module shall be capable of operating satisfactorily at the following loads for the designated operating cycles.

a. 10 A resistive load at 28 Vdc for 100,000 cycles.
b. 14 A resistive load at 32 Vdc for 25,000 cycles (any two contacts simultaneously but not the same pole).
c. 13 A, 0.2 percent power factor, inductive load at 115 V, 400 Hz, for 50,000 cycles (2 contacts in parallel).
d. 0.010 A resistive load at 28 Vdc for 100,000 cycles.

3.10.1 After life testing. After life testing the module shall conform to the end-of-life requirements of table I.

3.11 Dielectric withstanding voltage. Each module shall be capable of withstandi 1,000 plus or minus 50 Vac between all mutually insulated pins without mechanical or electrical degradation.

3.12 Preconditioning. Each module shall be subjected to a 5,000 cycle run-in test prior to shipment. This preconditioning may be conducted on the individual relay (NAVORD Drawing 2465232), or on the completed module and shall be performed as specifi in 4.4.3. When performed on the individual relay, the other components need not be preconditioned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall appl

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance wit procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptar tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (max per test figure)
Ohmmeter	Item R05	1
DC voltmeter	Item V03	1
DC power supply	Item P01	2
Electronic load	Item EL01	3
DC power supply	Item P06	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y01	2
Leakage tester, high voltage, AC	Item U02	1
Megohm bridge	Item R06	1
Resistors		
1.78 kΩ ±5%, 1/4 W		1
10 kΩ ±5%, 1/4 W		1
1 to 5 MΩ, 1/8 W		1
Transistor		
2N3108, or equivalent		1

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Switches	No. required (max per test figure)
SPST	1
SPDT	1

All resistors shall be metal film unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table 1.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved y the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) nd they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through L may be performed in any order.

4.4.3 Preconditioning. The 5,000 cycle run-in test shall be conducted under the following conditions:

a. Open circuit contact voltage: 6 ±1 V (Vac or Vdc).
b. Contact current: 10 ±2 mA (mAac or mAdc).
c. Cycling rate: 10 cycles per second maximum, on-off periods shall be equal within 10 percent.
d. Coil voltage: (pins 1A1(+) and 1A31(-) with pin 1B1 tied to 1A31); $8.4\ ^{+1.0}_{-0.0}$ Vdc.
e. Test equipment shall monitor the failure to close of each contact.

4.4.3.1 Contact resistance. A contact resistance greater than 0.500Ω for a closed contact during any cycle shall be cause to reject that module. Rejected modules shall not be shipped.

4.4.3.2 Preconditioning option. Relay preconditioning may be performed on the component level instead of the module level at the option of the manufacturer.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Sample size. The sample size shall be that required by MIL-M-28787, except that the life test shall require a minimum of three sample modules.

4.5.2 Test procedure. The test procedure shall be in accordance with MIL-M-28787, except as follows: (1) The overcurrent test shall be run on the module scheduled to go through the vibration, shock, mechanical, toxicity, and flammability tests, and shall be run after operating temperature tests have been completed on this module; (2) the overload tests shall be run on the module scheduled to go through thermal shock, humidity, and corrosion resistance tests, and shall be run after operating temperature tests have been completed on this module.

4.5.3 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through H, and L, in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through L shall be performed (except test condition J and K shall not be performed at 0°C) in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be nonoperating.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Coil resistance, K1	II	X	43.2	52.8	43.2	52.8	31.2	64.8	31.2	64.8	Ω
B1	Resistor, R3	III	X	817	833	782	868	800	850	765	885	Ω
B2	Resistor, R1	III	X	70.1	72.9	68.6	74.4	69.0	74.0	67.6	75.4	Ω
B3	Resistor, R2	III	X	44.7	49.3	43.7	50.3	43.9	50.1	43.0	51.0	Ω
B4	Resistor, R4	III	X	44.7	49.3	43.7	50.3	43.9	50.1	43.0	51.0	Ω
C1-C8	Contact resistance	IV	X	0.0	16	0.0	18	0.0	16	0.0	18	mV/A
D1-D61	Pin jumper continuity	V	S	0.0 (0.0)	25 (75)	0.0 (0.0)	25 (75)	0.0 (0.0)	25 (75)	0.0 (0.0)	25 (75)	mΩ (mV)
E1	Diode resistance (reverse), CR1	VI	X	10	∞	10	∞	10	∞	10	∞	MΩ
E2	Diode resistance (forward), CR1	VI	X	300	900	300	900	300	900	300	900	Ω
F	Dropout voltage	VII	S	0.5	-	0.5	-	0.5	-	0.5	-	Vdc
G	Operate time (including contact bounce)	VIII	X	0.0	9.0	0.0	9.0	0.0	10.0	0.0	10.0	ms
H	Release time (including contact bounce)	IX	S	0.0	15.0	0.0	15.0	0.0	15.0	0.0	15.0	ms
J1-J8	Dielectric strength @ ≤ 50,000 ft	X	S	← See note 3 →								

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TABLE I. Module electrical requirements - continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J9	Dielectric strength @ ≤ 50,000 ft, energized	X	S	← See note 3 →								
J10	Dielectric strength @ ≤ 50,000 ft, frame	X	S	← See note 3 →								
K1-K8	Insulation resistance	XI	S	100	∞	100	∞	100	∞	100	∞	MΩ
K9	Insulation resistance, energized	XI	S	100	∞	100	∞	100	∞	100	∞	MΩ
K10	Insulation resistance, frame	XI	X	100	∞	100	∞	100	∞	100	∞	MΩ
L	Pickup voltage	XII	S	0.0	6	0.0	6	0.0	6	0.0	8	Vdc

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. Test conditions J1-J10 shall have limits as specified in MIL-R-6106.

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4.5.3.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787.

4.5.4 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, closed contacts shall have a contact resistance of $\leq 0.1\Omega$ with 2 A $\pm 5\%$ passing through the contacts. The relay shall be energized during one-half of each test cycle and not energized during the other half. When energizing the relay coil, voltage shall be 25.0 $^{+3.0}_{-0.5}$ Vdc across pins A1(+) and B21(-) with pin B1 connected to pin A31.

4.5.5 Life tests. The life tests shall be conducted in accordance with MIL-M-28787 and the following. The module shall be operated as shown on figure 2. Prior to the performance of any measurements following the life tests, the insulation resistance test (test condition K) of table I shall be conducted. Any out of limit post life measurement shall be considered a relay failure.

4.5.5.1 Life test 1. All contacts shall be subjected to 100,000 cycles at a rate of 20 ± 2 cycles per minute, switching 10 $^{+1.0}_{-0.2}$ A resistive load at 28 $^{+3.0}_{-0.5}$ Vdc.

4.5.5.1.1 Life test 1 relay failure. A relay failure is defined as a maximum of four operations in which the dynamic resistance of a closed contact exceeds 0.500Ω. Following the test of 4.5.5.1, measure test conditions C, F, G, H, J, and K of table I. The measured values shall be as specified.

4.5.5.2 Life test 2. All contacts shall be subjected to 50,000 cycles at a rate of 20 ± 2 cycles per minute, switching 13 A inductive load with a 0.2 ± 10 percent power factor at 115 $^{+10}_{-1}$ Vac single phase only, at 400 ± 40 Hz (two contacts shall be connected in parallel for the test).

4.5.5.2.1 Life test 2 relay failure. A relay failure shall be defined as a maximum of two operations in which the dynamic resistance of a closed contact (or contacts in parallel) exceeds 0.500Ω. Following the test of 4.5.5.2, measure test conditions C, F, G, H, J, and K of table I. The measured values shall be as specified.

4.5.5.3 Life test 3. All contacts shall be subjected to 100,000 cycles at a rate of 1 to 10 cycles per second, switching 0.010 ± 0.001 A resistive load at 28 $^{+3.0}_{-0.5}$ Vdc.

4.5.5.3.1 Life test 3 relay failure. A relay failure shall be defined as a maximum of four operations in which the dynamic resistance of a closed contact exceeds 0.500Ω. Following the test of 4.5.5.3, measure test conditions C, F, G, H, J, and K of table I. The measured values shall be as specified except for test condition C which shall be 500 mΩ maximum.

4.5.5.4 Overload test. The units shall be tested in accordance with the requirement of MIL-R-6106. Each contact shall switch 20 $^{+2.0}_{-0.5}$ A (resistive load) at 28 $^{+3.0}_{-0.5}$ Vdc. Th duty cycle shall be 0.2 ± 0.05 seconds"on" time and 20 ± 1 seconds "off" time. The coil voltage shall be 25.0 $^{+3.0}_{-0.5}$ Vdc applied across pins A1(+) and B21(-) with pin B1 connecte to pin A31. The fin temperature shall be 60° $^{+5°}_{-0°}$ C. Following the test, measure test condition C of table I. Values shall be as specified.

4.5.5.5 Overcurrent test. The unit shall be energized with 3.5 ± 0.5 W of coil power at 60° $^{+5°}_{-0°}$ C (fin temperature) and each relay contact shall carry 10 $^{+1.0}_{-0.2}$ A at 28 $^{+3.0}_{-0.5}$ Vdc to a resistive load for a period of 72 hours minimum. Upon removal of the coil power after the 72 hour test, the relay release time shall be as specified in table I, test condition H.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance th MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to e end-of-life requirements of table I, the module is considered to have completed s useful life.

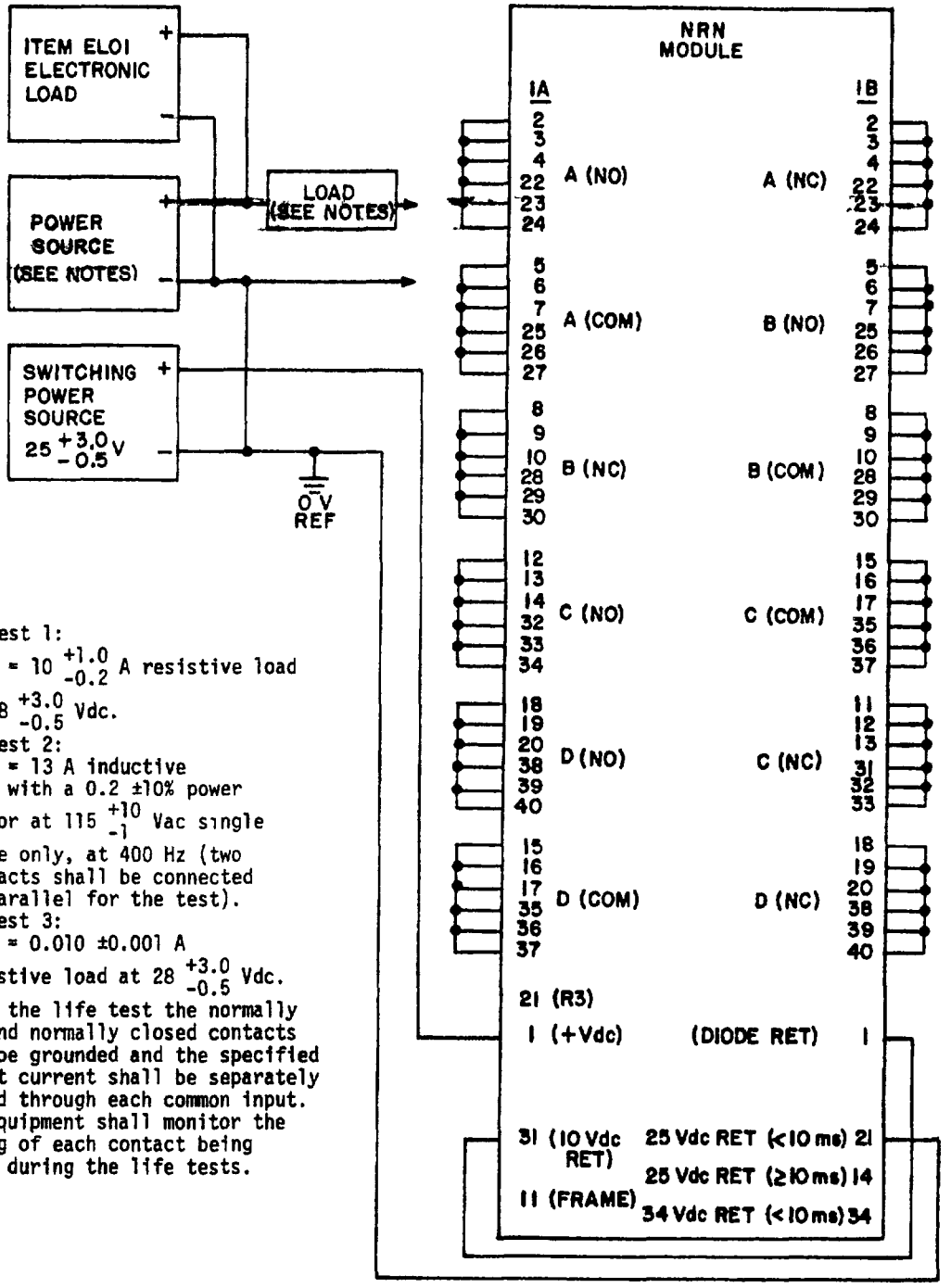
6.3 Changes from previous issue. The margins of this specification are marked with asterisk to indicate where changes (additions, modifications, corrections, deletions) om the previous issue were made. This was done as a convenience only and the vernment assumes no liability whatsoever for any inaccuracies in these notations. dders and contractors are cautioned to evaluate the marginal notations and relationship the last previous issue.

Preparing activity:
Navy - EC

Agent:
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NOTES:
1. Life test 1:
   Load = 10 $^{+1.0}_{-0.2}$ A resistive load at 28 $^{+3.0}_{-0.5}$ Vdc.
2. Life test 2:
   Load = 13 A inductive load with a 0.2 ±10% power factor at 115 $^{+10}_{-1}$ Vac single phase only, at 400 Hz (two contacts shall be connected in parallel for the test).
3. Life test 3:
   Load = 0.010 ±0.001 A resistive load at 28 $^{+3.0}_{-0.5}$ Vdc.
4. During the life test the normally open and normally closed contacts shall be grounded and the specified contact current shall be separately applied through each common input.
5. Test equipment shall monitor the closing of each contact being tested during the life tests.

FIGURE 2. Life test circuit.

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TABLE II. Coil resistance test.

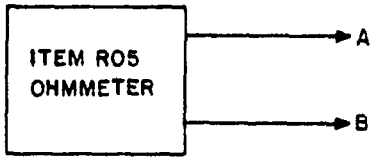
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	K1 Coil resistance	A1	A31	43.2	52.8	Ω	3

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TABLE III. Resistor test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	R3 Resistor	A1	A21	817	833	Ω	3
B2	R1	A31	B21	70.1	72.9		
B3	R2	B21	B14	44.7	49.3		
B4	R4 Resistor	B14	B34	44.7	49.3	Ω	3

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NRN MODULE

1A			1B
2			2
3			3
4			4
22	A (NO)	A (NC)	22
23			23
24			24
5			5
6			6
7			7
25	A (COM)	B (NO)	25
26			26
27			27
8			8
9			9
10			10
28	B (NC)	B (COM)	28
29			29
30			30
12			15
13			16
14			17
32	C (NO)	C (COM)	35
33			36
34			37
18			11
19			12
20			13
38	D (NO)	C (NC)	31
39			32
40		0	33
15			18
16			19
17			20
35	D (COM)	D (NC)	38
36			39
37			40
21	(R3)		
1	(+Vdc)	(DIODE RET)	1
31	(10 Vdc RET)	25 Vdc RET (<10 ms)	21
		25 Vdc RET (≥10 ms)	14
11	(FRAME)	34 Vdc RET (<10 ms)	34

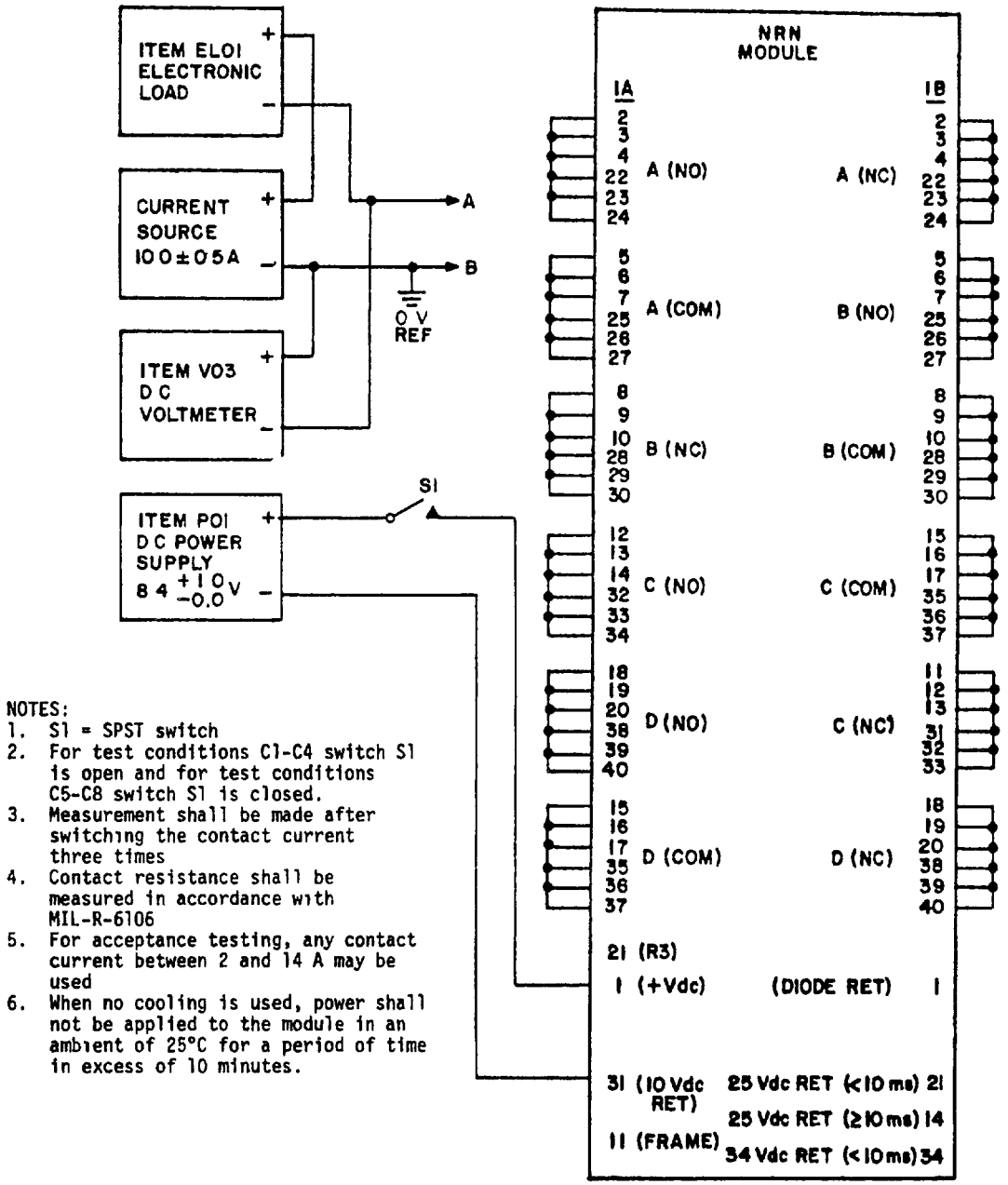
FIGURE 3. Coil resistance and resistor test circuit for tables II and III.

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TABLE IV. Contact resistance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	NC Contact resistance	A5	B2	0.0 (0.0)	16 (168)	mΩ (mV)	4
C2	NC	B8	A8				
C3	NC	B15	B11				
C4	NC	A15	B18				
C5	NO	A2	A5				
C6	NO	B5	B8				
C7	NO	A12	B15				
C8	NO Contact resistance	A18	A15	0.0 (0.0)	16 (168)	mΩ (mV)	4

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NOTES:
1. S1 = SPST switch
2. For test conditions C1-C4 switch S1 is open and for test conditions C5-C8 switch S1 is closed.
3. Measurement shall be made after switching the contact current three times
4. Contact resistance shall be measured in accordance with MIL-R-6106
5. For acceptance testing, any contact current between 2 and 14 A may be used
6. When no cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes.

FIGURE 4 Contact resistance test circuit for table IV.

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TABLE V. Pin jumper continuity test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	Pin jumper continuity	A2	A3	Open	Open	0.0 (0.0)	25 (75)	mΩ (mV)	5
D2			A4						
D3			A22						
D4			A23						
D5		A2	A24						
D6		A5	A6						
D7			A7						
D8			A25						
D9			A26						
D10		A5	A27						
D11		B2	B3						
D12			B4						
D13			B22						
D14			B23						
D15		B2	B24						
D16		B5	B6						
D17		B5	B7						
D18	Pin jumper continuity	B5	B25	Open	Open	0.0 (0.0)	25 (75)	mΩ (mV)	5

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TABLE V. Pin jumper continuity test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D19	Pin jumper continuity	B5	B26	Open	Open	0.0 (0.0)	25 (75)	mΩ (mV)	5
D20		B5	B27						
D21		B8	B9						
D22			B10						
D23			B28						
D24			B29						
D25		B8	B30						
D26		A8	A9						
D27			A10						
D28			A28						
D29			A29						
D30		A8	A30						
D31		A12	A13						
D32			A14						
D33			A32						
D34			A33						
D35		A12	A34						
D36	Pin jumper continuity	B15	B16	Open	Open	0.0 (0.0)	25 (75)	mΩ (mV)	5

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TABLE V. Pin jumper continuity test - Continued

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D37	Pin jumper continuity	B15	B17	Open	Open	0.0 (0.0)	25 (75)	mΩ (mV)	5
D38			B35						
D39			B36						
D40		B15	B37						
D41		B11	B12						
D42			B13						
D43			B31						
D44			B32						
D45		B11	B33						
D46		A18	A19						
D47			A20						
D48			A38						
D49			A39						
D50		A18	A40						
D51		A15	A16						
D52			A17						
D53			A35						
D54	Pin jumper continuity	A15	A36	Open	Open	0.0 (0.0)	25 (75)	mΩ (mV)	5

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TABLE V. Pin jumper continuity test - Continued.

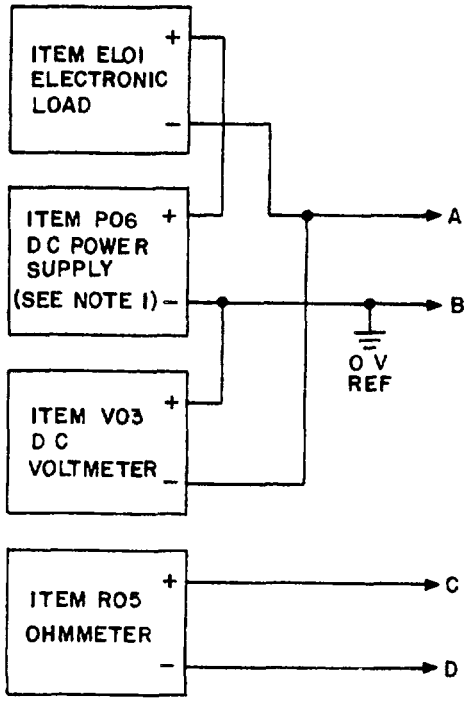
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D55	Pin jumper continuity	A15	A37	Open	Open	0.0 (0.0)	25 (75)	mΩ (mV)	5
D56		B18	B19						
D57			B20						
D58			B38						
D59			B39						
D60		B18	B40						
D61	Pin jumper continuity	A11	Frame	Open	Open	0.0 (0.0)	25 (75)	mΩ (mV)	5

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TABLE VI. Diode resistance (equivalent) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	CR1 Diode resistance (reverse)	Open	Open	A1	B1	10	∞	MΩ	5
E2	CR1 Diode resistance (forward)	Open	Open	B1	A1	300	900	Ω	5

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NOTES:
1. Adjust item P06 for 3.00 ±0.03 A at the module pin.
2. When no cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes.
3. Test conditions E1 and E2 shall be measured using a current of 1 mA maximum.

NRN MODULE

1A			1B
2			2
3			3
4			4
22	A (NO)	A (NC)	22
23			23
24			24
5			5
6			6
7			7
25	A (COM)	B (NO)	25
26			26
27			27
8			8
9			9
10			10
28	B (NC)	B (COM)	28
29			29
30			30
12			15
13			16
14			17
32	C (NO)	C (COM)	35
33			36
34			37
18			11
19			12
20			13
38	D (NO)	C (NC)	31
39			32
40			33
15			18
16			19
17			20
35	D (COM)	D (NC)	38
36			39
37			40
21	(R3)		
1	(+Vdc)	(DIODE RET)	1
31	(10 Vdc RET)	25 Vdc RET (<10 ms)	21
		25 Vdc RET (≥10 ms)	14
11	(FRAME)	34 Vdc RET (<10 ms)	34

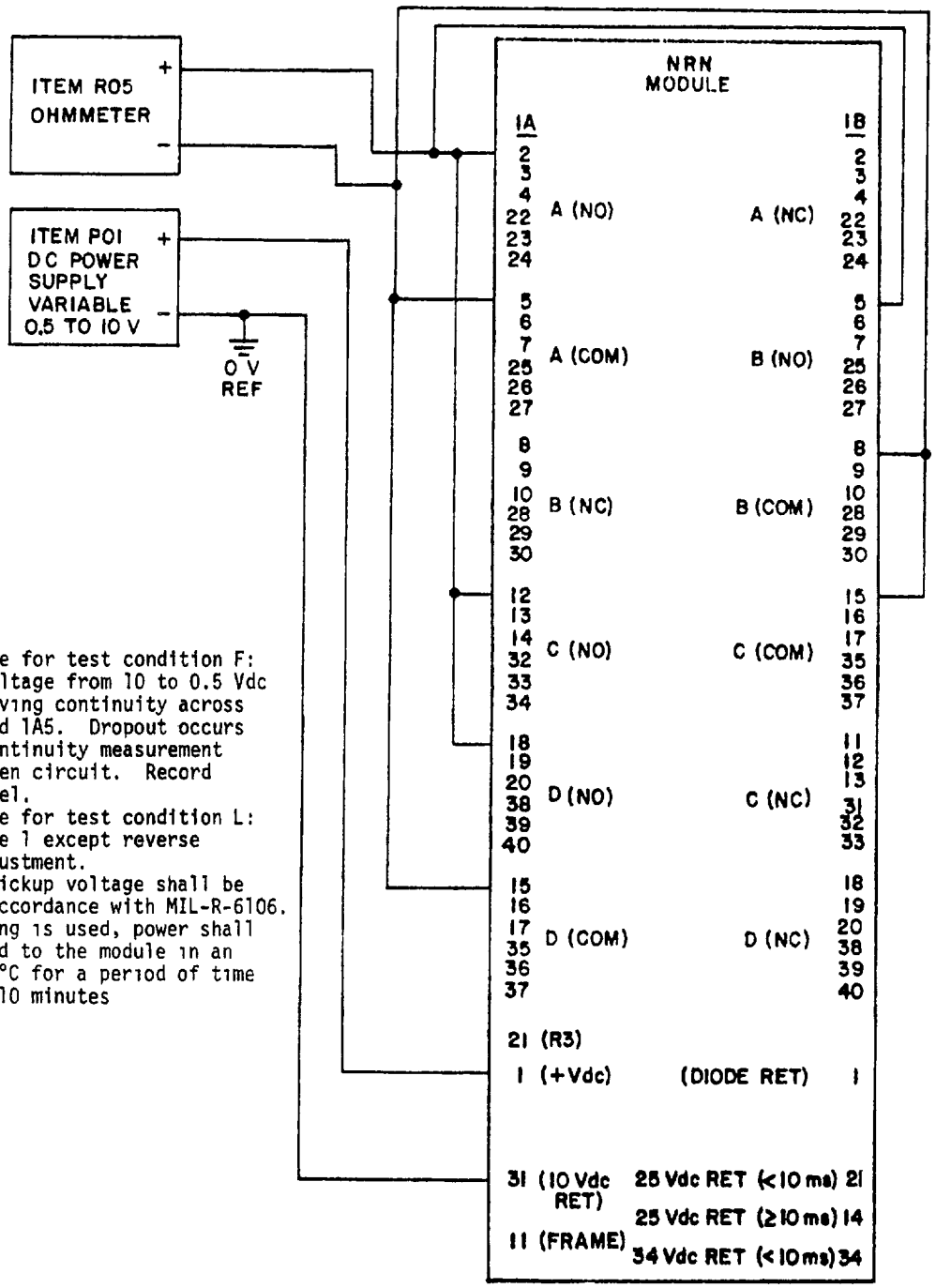
FIGURE 5. Pin jumper continuity and diode resistance test circuit for tables V and VI

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TABLE VII. Dropout voltage test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
F	Dropout voltage	0.5	---	Vdc	6

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ES:
Test procedure for test condition F:
Decrease voltage from 10 to 0.5 Vdc while observing continuity across pins 1A2 and 1A5. Dropout occurs when the continuity measurement shows an open circuit. Record voltage level.
Test procedure for test condition L:
Same as note 1 except reverse voltage adjustment.
Dropout and pickup voltage shall be measured in accordance with MIL-R-6106.
When no cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes

FIGURE 6. Dropout and pickup voltage test circuit for tables VII and XII.

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TABLE VIII. Operate time (including contact bounce) test.

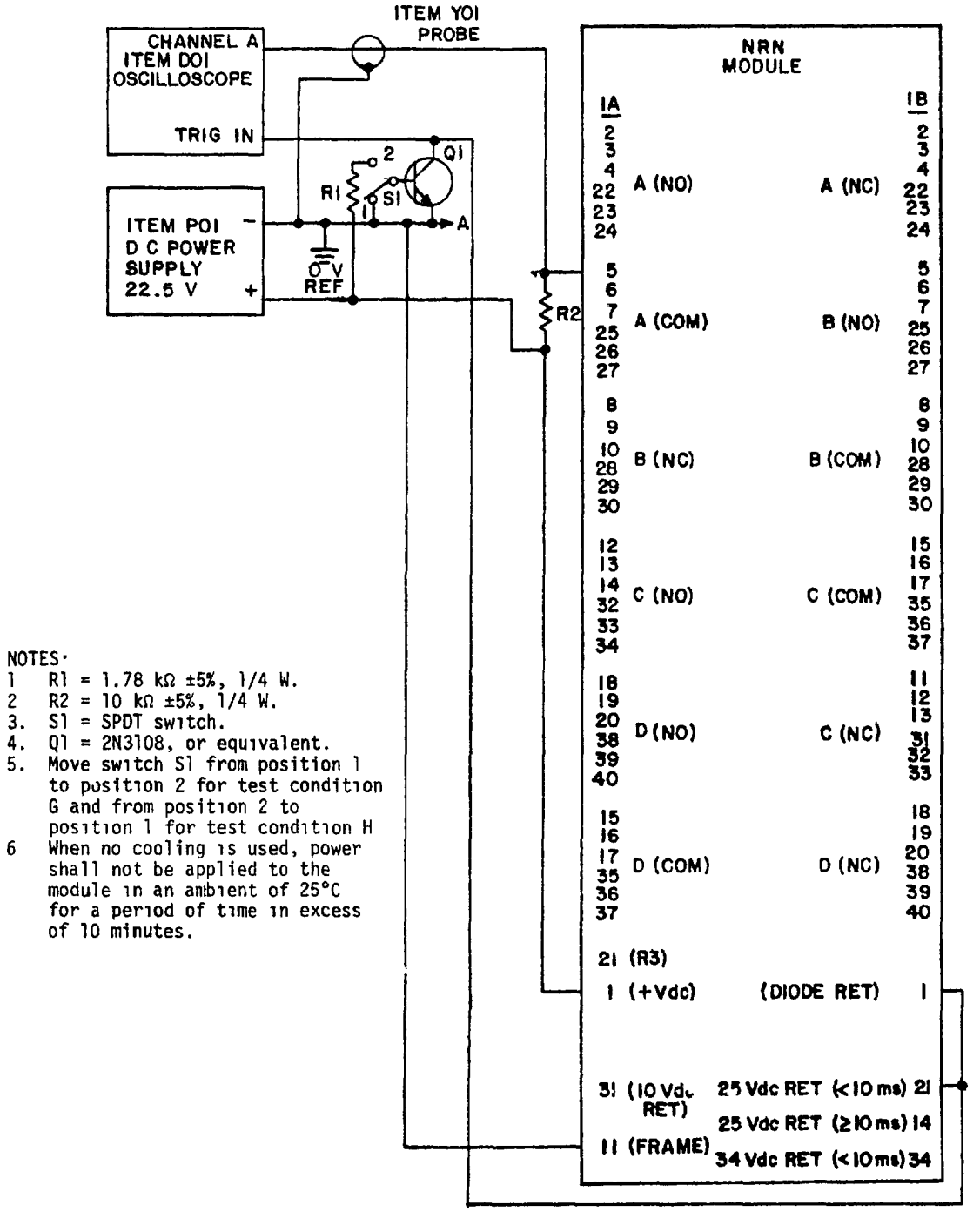
Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G	Operate time (including contact bounce)	A2	0.0	9.0	ms	7

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TABLE IX. Release time (including contact bounce) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
H	Release time (including contact bounce)	B2	0.0	15.0	ms	7

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NOTES:
1. R1 = 1.78 kΩ ±5%, 1/4 W.
2. R2 = 10 kΩ ±5%, 1/4 W.
3. S1 = SPDT switch.
4. Q1 = 2N3108, or equivalent.
5. Move switch S1 from position 1 to position 2 for test condition G and from position 2 to position 1 for test condition H
6. When no cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes.

FIGURE 7 Operate time (including contact bounce) and release time (including contact bounce) test circuit for tables VIII and IX

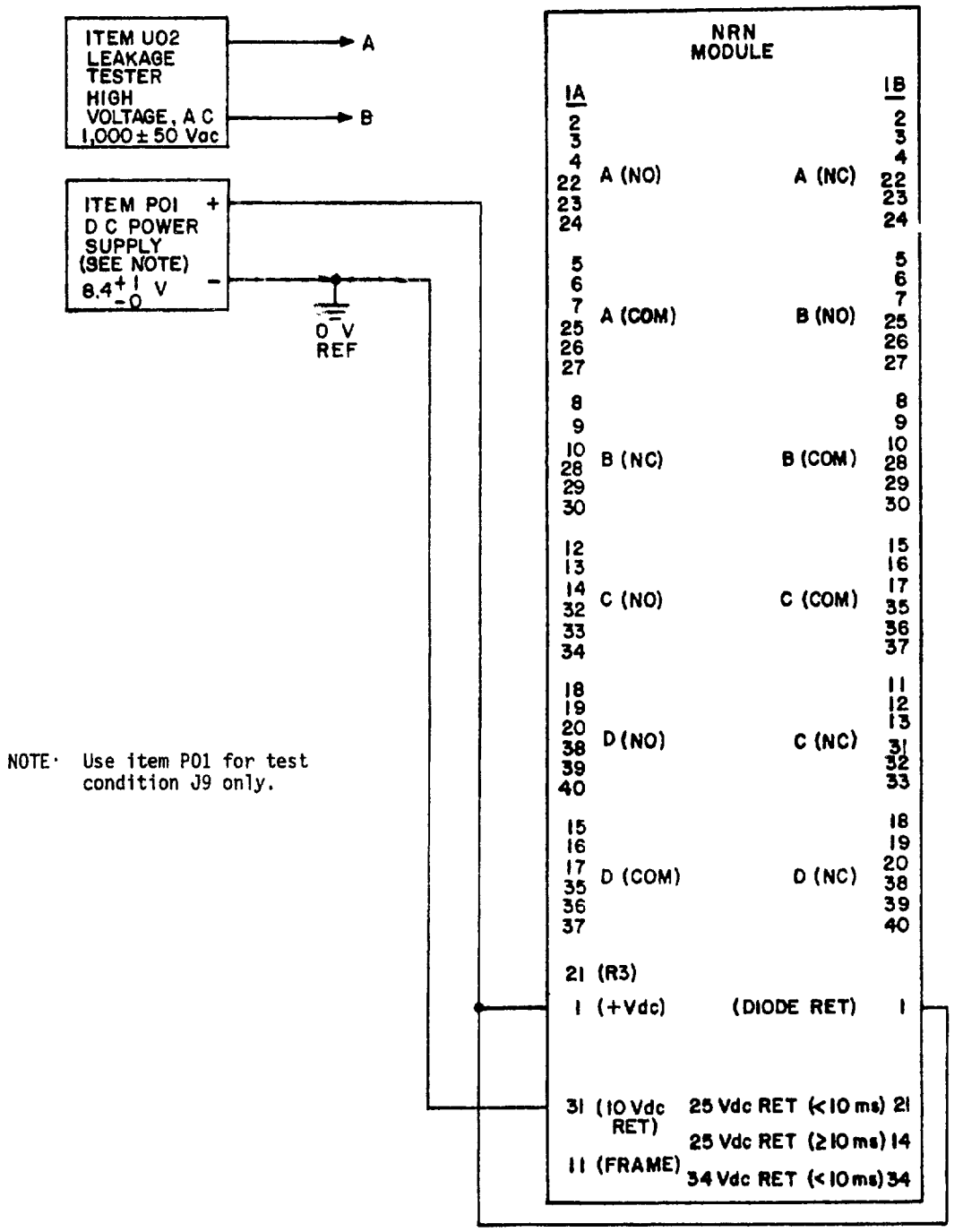
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TABLE X. Dielectric strength test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin (See note 3)	B to pin	Min	Max		
J1	Dielectric strength @ ≤ 50,000 ft	Coil, A(NO)	All others	See notes 1 and 2			8
J2		Coil, A(NC), A(COM)					
J3		Coil, B(NO)					
J4		Coil, B(NC), B(COM)					
J5		Coil, C(NO)					
J6		Coil, C(NC), C(COM)					
J7		Coil, D(NO)					
J8	Dielectric strength @ ≤ 50,000 ft	Coil, D(NC), D(COM)	All others				
J9	Dielectric strength @ ≤ 50,000 ft, energized	A(COM), A(NO), B(COM), B(NO), C(COM), C(NO), D(COM), D(NO)	A(NC), B(NC), C(NC), D(NC)				
J10	Dielectric strength @ ≤ 50,000 ft, frame	1All	All others	See notes 1 and 2			8

NOTES:
1. When a relay is tested as specified, any arc-over (air discharge), flash-over (surface discharge), or breakdown (puncture discharge) causing a leakage current of more than 1.0 mA at the specified test voltage shall constitute a failure.
2. For the qualification test, modules shall be subjected to a fin temperature of 60° $^{+5°}_{-0°}$ C. For acceptance tests, modules shall be subjected to a fin temperature of 25° ±5°C.
3. Coil = 1A1, 1A21, 1A31, 1B1, 1B14, 1B21, and 1B34.

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NOTE: Use item P01 for test condition J9 only.

FIGURE 8. Dielectric strength test circuit for table X.

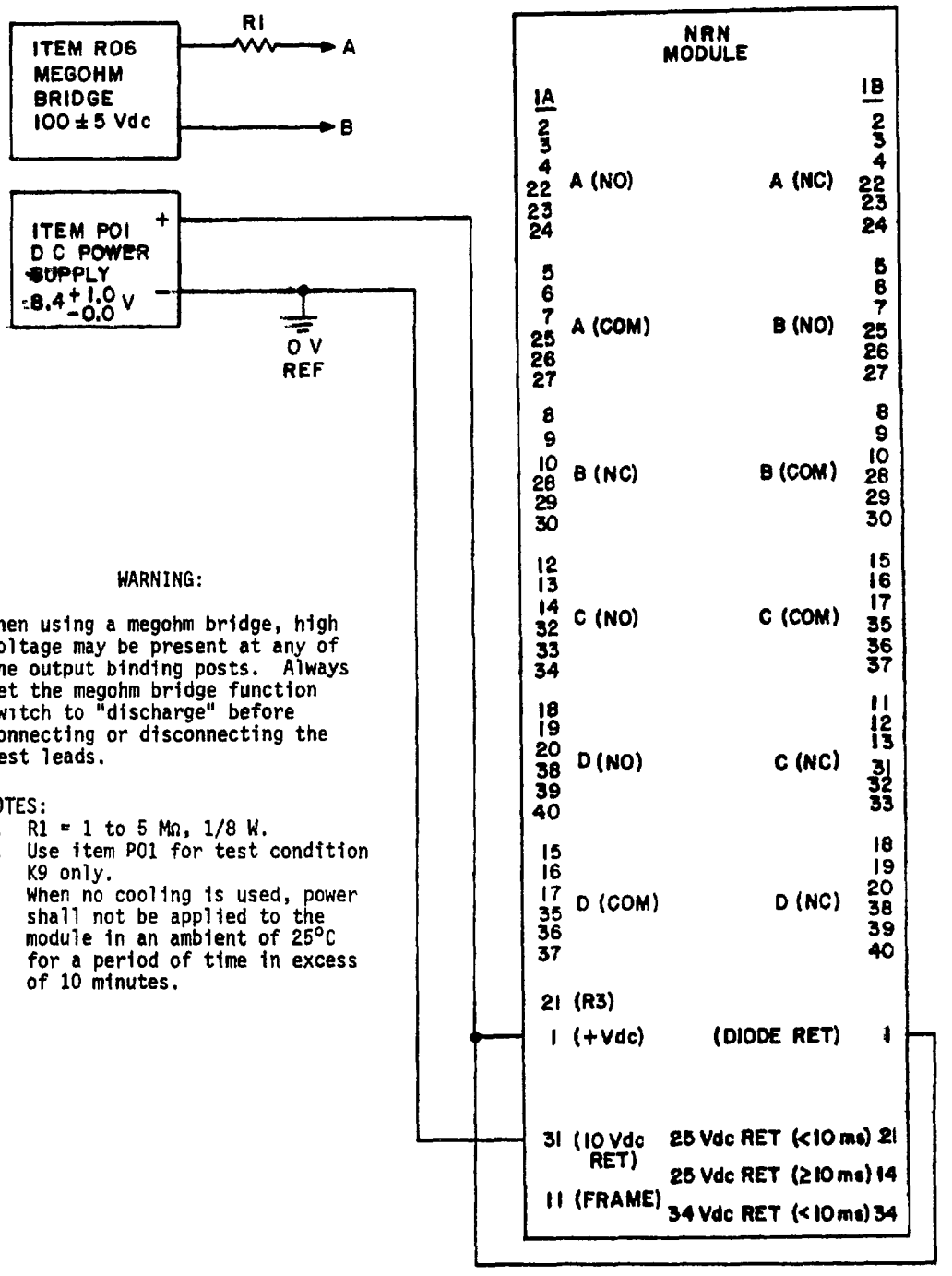
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TABLE XI. Insulation resistance test.

Test condition	Parameter	Connect		Initial limits (See notes 1 and 2)		Unit	Test figure
		A to pin (See note 3)	B to pin	Min	Max		
K1	Insulation resistance	Coil, A(NO)	All others	100	∞	MΩ	9
K2	↓	Coil, A(NC), A(COM)	↓	↓	↓	↓	↓
K3		Coil, B(NO)					
K4		Coil, B(NC), B(COM)					
K5		Coil, C(NO)					
K6		Coil, C(NC), C(COM)					
K7		Coil, D(NO)					
K8	Insulation resistance	Coil, D(NC), D(COM)	All others				↓
K9	Insulation resistance, energized	A(COM), A(NO), B(COM), B(NO), C(COM), C(NO), D(COM), D(NO)	A(NC), B(NC), C(NC), D(NC)				9
K10	Insulation resistance, frame	1A11	All others	100	∞	MΩ	10

NOTES:
1. For test conditions K1 through K9, the module shall be subjected to an operating temperature of 60° $^{+5°}_{-0°}$ C.
2. For test condition K10, the module shall be subjected to an operating temperature of 25° ±5°C.
3. Coil = 1A1, 1A21, 1A31, 1B1, 1B14, 1B21, and 1B34.

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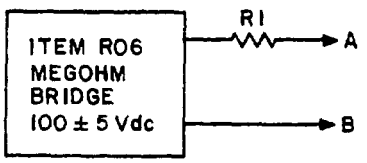
WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. R1 = 1 to 5 MΩ, 1/8 W.
2. Use item P01 for test condition K9 only.
3. When no cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes.

FIGURE 9. Insulation resistance test circuit for table XI (test conditions K1 through K9).

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

NOTE: R1 = 10 kΩ ±5%, 1/4 W.

NRN MODULE			
1A			1B
2			2
3			3
4			4
22	A (NO)	A (NC)	22
23			23
24			24
5			5
6			6
7			7
25	A (COM)	B (NO)	25
26			26
27			27
8			8
9			9
10			10
28	B (NC)	B (COM)	28
29			29
30			30
12			15
13			16
14			17
32	C (NO)	C (COM)	35
33			36
34			37
18			11
19			12
20			13
38	D (NO)	C (NC)	31
39			32
40			33
15			18
16			19
17			20
35	D (COM)	D (NC)	38
36			39
37			40
21	(R3)		
1	(+Vdc)	(DIODE RET)	1
31	(10 Vdc RET)	25 Vdc RET (<10 ms)	21
		25 Vdc RET (≥10 ms)	14
11	(FRAME)	34 Vdc RET (<10 ms)	34

FIGURE 10. Insulation resistance test circuit for table XI (test condition K10).

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Test condition	Parameter	Initial limits 25° +5°C Ambient		Unit	Test figure
		Min	Max		
L	Pickup voltage	0.0	6	Vdc	6

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